

STRUCTURAL DESIGN CRITERIA

1. BUILDING CODE:
2018 INTERNATIONAL RESIDENTIAL CODE

2. GRAVITY LOADS:

ROOF

CONCENTRATED

LL = N/A
DL = SEE PLANS

DISTRIBUTED

LL = 20 PSF
DL = 15 PSF

ELEVATED FLOOR

LL = 2000 LBS
DL = 150 LBS

LL = SEE CHART
DL = 20 PSF

1ST FLOOR

LL = 2000 LBS
DL = 150 LBS

LL = SEE CHART
DL = 50 PSF

COLLATERAL LOAD

SEE PLANS

CL = 5 PSF

3. WIND LOADS:

ULTIMATE DESIGN WIND SPEED, V_{100} = 140mph
NOMINAL DESIGN WIND SPEED, V_{50} = 108.4mph
RISK CATEGORY II
WIND EXPOSURE B
INTERNAL PRESSURE COEFFICIENT, GC_{pi} = ± 0.18

4. SEISMIC LOADS:

RISK CATEGORY II
SEISMIC IMPORTANCE FACTOR, I_e = 1.00
MAPPED SPECTRAL RESPONSE ACCELERATION PARAMETERS:
 S_{D1} = 0.404g; S_{D2} = 0.144g
SITE CLASS D
DESIGN SPECTRAL RESPONSE ACCELERATION PARAMETERS:
 S_{S1} = 0.389g; S_{S2} = 0.214g
SEISMIC DESIGN CATEGORY D
BASIC SEISMIC FORCE-RESISTING SYSTEM:
LIGHT FRAMED WALLS W/ WOOD SHEAR PANELS
DESIGN BASE SHEAR:
 V = CS
 W = EFFECTIVE SEISMIC MASS
SEISMIC RESPONSE COEFFICIENT, C_s = 0.07
RESPONSE MODIFICATION FACTOR, R = 6.5
ANALYSIS PROCEDURE:
EQUIVALENT LATERAL FORCE METHOD

5. SNOW LOAD:
GROUND SNOW LOAD P_g = 0 PSF

RESIDENTIAL LIVE LOAD CHART

OCCUPANCY

UNIFORMLY DISTRIBUTED LOAD

ATTICS W/ LIMITED STORAGE

20psf

ATTICS W/O STORAGE

10psf

DECKS

40psf

EXTERNAL BALCONIES

60psf

FIRE ESCAPES

40psf

GUARDRAILS & HANDRAILS

200psf

GUARDRAIL INFILL COMPONENTS

50psf

PASSENGER VEHICLE GARAGES

50psf

ROOMS OTHER THAN SLEEPING ROOMS

40psf

SLEEPING ROOMS

30psf

STAIRS

40psf

ZONE 2

ZONE 3

ZONE 5

ZONE 5

ZONE 5

HIP ROOF

GABLE ROOF

$\alpha = 3.0$ FT

COMPONENTS & CLADDING

WIND PRESSURES (PSF)

ZONE

AREA (SF)

+P

-P

ROOF > 7° TO 27°

10

20.3

32.3

20

18.5

31.4

30

16.1

30.2

100

16.0

29.1

10

20.3

55.2

20

18.5

51.7

30

16.1

45.7

100

16.0

41.2

10

20.3

63.1

20

18.5

77.7

30

16.1

70.5

100

16.0

65.1

10

35.3

36.2

20

33.7

36.7

50

31.6

34.6

100

33.0

33.0

10

35.3

47.2

20

33.7

44.0

50

31.6

39.8

100

33.0

36.7

WALLS

10

35.3

36.2

20

33.7

36.7

50

31.6

34.6

100

33.0

33.0

10

35.3

47.2

20

33.7

44.0

50

31.6

39.8

100

33.0

36.7

FRAMING SPECIFICATIONS

1) GENERAL

1.1) IN ADDITION TO THE CONNECTION AND FASTENING REQUIREMENTS SHOWN AND SPECIFIED ON THESE DRAWINGS, ALL FRAMING CONNECTIONS SHALL MEET OR EXCEED THE REQUIREMENTS SET FORTH IN TABLE 2306.1 OF THE IRC 2018.

1.2) UNLESS NOTED OTHERWISE ON THE DRAWINGS, ALL FRAMING LUMBER SHALL BE GROUP II SPECIES WITH A SPECIFIC GRAVITY OF 0.49 OR GREATER, (SOUTHERN PINE, DOUGLAS FIR etc.)

2) FLOOR SYSTEMS

2.1) SHEATHING: ALL SHALL BE A MINIMUM OF 15/32" THICK WOOD STRUCTURAL PANELS INSTALLED LONG DIMENSION PERPENDICULAR TO FRAMING MEMBERS. END JOINTS SHALL BE STAGGERED OVER COMMON FRAMING MEMBERS

2.2) CONNECTIONS: FOR JOISTS SUBJECT TO WIND UPLIFT LOADS, CONNECTIONS TO SILL, GIRDER OR BAND BEAM SHALL BE MADE WITH SIMPSON FACE MOUNT HANGERS LUS28 FOR 2x8 JOISTS AND LUS28 FOR 2x10 AND 2x12 JOISTS. THESE AREAS ARE DENOTED ON THE DRAWINGS. CONNECTIONS OF JOIST FLOOR SYSTEMS SHALL BE SPECIFIED BY JOIST SUPPLIER.

2.3) BRACING: PROVIDE FULL DEPTH BLOCKING PERPENDICULAR TO THE FLOOR FRAMING MEMBERS IN THE FIRST TWO FRAMING SPACES AT EACH END OF THE FLOOR SYSTEM, SPACED 4' MAX o.c.

2.4) FASTENING: FLOOR SHEATHING SHALL BE FASTENED TO FRAMING MEMBERS AS FOLLOWS: 8' CEILINGS FIRST AND SECOND FLOOR AND 10' CEILINGS FIRST FLOOR, SHEATHING SHALL BE FASTENED TO FRAMING MEMBERS WITH 8d NAILS @ 8" o.c. ALONG ALL EDGES AND 12" o.c. IN THE FIELD. BLOCKING OF FRAMING MEMBERS IS NOT REQUIRED. 10' CEILING HEIGHT SECOND FLOOR, SHEATHING SHALL BE FASTENED TO FRAMING MEMBERS WITH 8d NAILS @ 8" o.c. ALONG ALL EDGES AND 12" o.c. IN THE FIELD. BLOCKING OF FRAMING MEMBERS IS REQUIRED.

3) WALL SYSTEMS

3.1) STUDS: STUD SIZES, GRADE, SPACING AND LENGTH SHALL CONFORM TO THE FOLLOWING: 8' AND 10' EXTERIOR LOADBEARING WALLS WITH MIN 3/8" EXT SHEATHING AND NO INTERIOR SHEATHING SHALL BE 2x4 @ 16" o.c. OF EITHER GROUP II SPECIES GRADES SS #1, #2 AND CONST. OR SPRUCE-PINE-FIR GRADES SS #1, #2, 8' AND 10' EXTERIOR LOADBEARING WALLS WITH MIN 3/8" EXT SHEATHING AND 1/2" GYP BOARD INTERIOR SHEATHING SHALL BE 2x4 @ 16" o.c. OF EITHER GROUP II SPECIES GRADES SS #1, #2 AND CONST. OR SPRUCE-PINE-FIR GRADES SS #1, #2 CONST. EXTERIOR NON-LOADBEARING WALLS (GABLE END) FOR WALLS 10' OR LESS, USE SAME SPEC FOR LOADBEARING WALLS.

3.2) HEADERS: HEADERS AND OPENINGS SHALL CONFORM TO THE DETAILS AND SCHEDULES SHOWN ON THE DRAWINGS.

3.3) CONNECTIONS: CONNECTIONS FOR EXTERIOR WALL FRAMING SHALL CONFORM TO DETAILS AND HURRICANE TIE SCHEDULE SHOWN ON THE DRAWINGS.

3.4) TOP PLATE SPLICE: ALL SPLICES IN THE WALL TOP PLATES SHALL CONFORM TO THE DETAIL SHOWN ON THE DRAWINGS

4) SHEARWALLS

4.1) GENERAL: SHEARWALLS CAN BE EITHER TYPE I OR TYPE II SHEARWALLS. SEE SHEARWALL PLAN FOR TYPE DEFINITION. TYPE I SHEARWALLS ARE SEGMENTED, WITH EACH SEGMENT SHEATHED WITH FULL HEIGHT SHEATHING. A HOLDOWN SHALL BE LOCATED AT EACH END OF EACH SEGMENT. STUDS SHALL BE DOUBLED AT THE ENDS OF SHEARWALL SEGMENTS AND THE SEGMENTS SHALL BE CONNECTED BY A DOUBLE TOP PLATE DRAG STRUT.

4.2) FULL HEIGHT SHEATHING: WITH THE EXCEPTION OF THE BUILDING CORNERS, THE MINIMUM LENGTH OF THE SHEATHING SHALL BE 2'-6" FOR 8' WALLS AND 3'-0" FOR 10' WALLS. BUILDING CORNERS SHALL BE SHEATHED FOR A MINIMUM OF 27" FOR 8' WALLS AND 34" FOR 10' WALLS. OPENINGS SHALL BE LIMITED TO 144 sq in WITH A MAXIMUM DIMENSION OF 12". ALL OPENINGS SHALL BE BLOCKED.

4.3) SHEATHING: ALL SHEATHING SHALL BE 7/16" MINIMUM WOOD STRUCTURAL PANELS NAILED IN ACCORDANCE WITH THE SHEARWALL SCHEDULE SHOWN ON THE DRAWINGS.

4.4) FASTENING: ALL SHEATHING SHALL BE FASTENED TO FRAMING MEMBERS ON ALL FOUR SIDES. ALL EDGES SHALL BE BLOCKED. JOINTS FOR ADJACENT PANELS SHALL OCCUR OVER COMMON FRAMING MEMBERS OR BLOCKING. WHERE PANEL EDGES ABUT, FRAMING AND BLOCKING SHALL BE A MINIMUM OF 3" NOMINAL. NAILS SHALL BE STAGGERED. NAILS IN PANEL FIELD SHALL BE 12" o.c. MAXIMUM. EDGE NAIL SPACING AND NAIL SIZE ARE SPECIFIED ON THE DRAWINGS SHEATHING SHALL BE NAILED TO THE TOP MEMBER OF THE TOP PLATE.

5) ROOF SYSTEMS

5.1) RAFTER-JOIST SYSTEMS: SPACING SHALL BE 16" o.c. PROVIDE A MINIMUM 2" RIDGE BOARD WITH A DEPTH NOT LESS THAN THE DEPTH OF THE RAFTER. RAFTERS SHALL BE PLACED DIRECTLY OPPOSITE EACH OTHER AND BEAR AGAINST THE RIDGE BOARD. INSTALL A 2x6 COLLAR BEAM IN THE UPPER THIRD OF THE ROOF TO EVERY THIRD PAIR OF RAFTERS.

5.2) RAFTER SIZES SHALL CONFORM TO THE FOLLOWING:
UNSUPPORTED RAFTER SPAN UP TO 6'-6" - 2X4 MINIMUM
UNSUPPORTED RAFTER SPAN UP TO 10'-0" - 2X6 MINIMUM
UNSUPPORTED RAFTER SPAN UP TO 13'-6" - 2X8 MINIMUM
UNSUPPORTED RAFTER SPAN UP TO 17'-6" - 2X10 MINIMUM

5.3) EXTERIOR WALL CONNECTIONS ARE SHOWN ON THE DRAWINGS.
INTERIOR CONNECTIONS ARE TO CONFORM TO THE FOLLOWING REQUIREMENTS
UNSUPPORTED RAFTER SPAN UP TO 12"
UPLIFT = 250#. SHEAR PERPENDICULAR TO RAFTER = 280#
UNSUPPORTED RAFTER SPAN UP TO 16"
UPLIFT = 351#. SHEAR PERPENDICULAR TO RAFTER = 280#
UNSUPPORTED RAFTER SPAN UP TO 20"
UPLIFT = 369#. SHEAR PERPENDICULAR TO RAFTER = 280#

5.4) HIP AND VALLEY RAFTER SIZES AND CONNECTIONS SHALL CONFORM TO THE FOLLOWING REQUIREMENTS:
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 6', MINIMUM SIZE (1) 2x6
HIGH END UPLIFT = 150# LOW END UPLIFT = 109#
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 8', MINIMUM SIZE (1) 2x6
HIGH END UPLIFT = 250# LOW END UPLIFT = 175#
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 10', MINIMUM SIZE (1) 2x6
HIGH END UPLIFT = 409# LOW END UPLIFT = 275#
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 12', MINIMUM SIZE (2) 2x8 OR (1) 2X10 OR 1.75x5.5 LVL
HIGH END UPLIFT = 580# LOW END UPLIFT = 375#
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 14', MINIMUM SIZE (2) 2x8 OR (1) 2x12 OR (1) 1.75x7.25 LVL
HIGH END UPLIFT = 840# LOW END UPLIFT = 540#
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 16', MINIMUM SIZE (2) 2x10 OR (1) 1.75x9.25 LVL
HIGH END UPLIFT = 1120# LOW END UPLIFT = 720#
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 18', MINIMUM SIZE (2) 2x12 OR (1) 1.75x9.5 LVL
HIGH END UPLIFT = 1450# LOW END UPLIFT = 920#
UNSUPPORTED HORIZONTAL RAFTER SPAN UP TO 20', MINIMUM SIZE (3) 2x12 OR (1) 1.75x11.25 LVL
HIGH END UPLIFT = 1800# LOW END UPLIFT = 1140#

5.5) BRACING: WHEN A GABLE ENDWALL EXTENDS FROM THE TOP FLOOR TO THE ROOF SHEATHING AND IS NOT SUPPORTED BY A CEILING DIAPHRAGM, END WALL ROOF BRACING SHALL BE PROVIDED. THIS SHALL CONSIST OF 2x BRACING MEMBERS THE FULL DEPTH OF THE RAFTER OR TRUSS TOP CHORD INSTALLED IN THE FIRST TWO END SPACES PERPENDICULARS TO THE ROOF FRAMING SYSTEM. BRACING SHALL BE PLACED AT 4' o.c. MAX. THE ROOF SHEATHING SHALL BE NAILED TO THE BRACING WITH THE PANEL NAILING PROCEDURE SET FORTH IN 5.7 OF THIS SPECIFICATION. SEE ENDWALL BRACING DIAGRAM ON THE DRAWINGS.

5.6) ROOF SHEATHING: ALL SHALL BE A MINIMUM THICKNESS OF 15/32" EXPOSURE I WOOD STRUCTURAL PANELS INSTALLED IN ACCORDANCE WITH THE ROOF SHEATHING LAYOUT ON THE DRAWINGS. PANEL LONG DIMENSION SHALL BE PERPENDICULAR TO FRAMING AND END JOINTS SHALL BE STAGGERED.

6) HOLDOWNS

6.1) GENERAL: HOLDOWNS SHALL BE REQUIRED ON ALL INTERIOR AND EXTERIOR SHEAR WALLS. HOLDOWNS SHALL BE LOCATED AT EACH END OF EACH SEGMENT OF TYPE I SHEARWALLS. HOLDOWNS SHALL BE LOCATED AT EACH END OF SHEARWALLS FOR TYPE II WALLS UNLESS NOTED OTHERWISE ON THE DRAWINGS AT BUILDING CORNERS. UNDER CERTAIN CONDITIONS, A SINGLE HOLDOWN MAY BE PERMITTED AT BUILDING CORNERS WHERE TWO SHEARWALLS OR TWO SHEARWALL SEGMENTS MEET. SPECIAL NAILING REQUIREMENTS SHALL BE PROVIDED FOR THIS CONDITION. SEE DRAWINGS.

6.2) HOLDOWNS: ALL SHALL BE GALVANIZED TYPE HD, HTT AND LTT MANUFACTURED BY SIMPSON. EQUIVALENT USP HOLDOWNS MAY BE SUBSTITUTED. SEE DRAWINGS FOR HOLDOWN IDENTIFICATION AND INSTALLATION DETAILS.

7) METAL CONNECTORS

7.1) THE METAL CONNECTORS SHOWN ON THE DRAWINGS ARE SPECIFIED AS EITHER SIMPSON OR USP. THESE TWO MANUFACTURER'S CONNECTORS MAY BE SUBSTITUTED FOR ONE ANOTHER AS LONG AS THE SUBSTITUTION IS EQUAL TO OR GREATER IN LOAD CAPACITY.

EXT. SHEATHING NAILING SPECIFICATION:

WALL SHEATHING 7/16" OSB OR 15/32" PLYWOOD:
FASTEN SHEATHING W/ 2-3/8" X 0.113" DIA. NAILS@ 3" O.C. AT EDGES & 12" O.C. IN THE PANEL FIELD.
(TYP. U.N.O.)

ALL SHEATHING PANELS SHALL BE ORIENTED AND INSTALLED FULL HEIGHT OF SHEAR WALL OR 2X HORIZONTAL
BLOCKING SHALL BE PROVIDED TO SUPPORT ALL UNSUPPORTED EDGES & EDGE FASTENING.

ALL EXTERIOR WALLS SHALL BE CONTINUOUSLY SHEATHED AND ARE CONSIDERED SHEAR WALLS.

ROOF SHEATHING SPECIFICATION:

ROOF SHEATHING SHALL BE 7/16" A.P.A. RATED SHEATHING 24/16 EXPOSURE 1 (OR APPROVED EQUAL).

FASTEN SHEATHING:
W/ 2-1/2" X 0.131" DIA. NAILS@ 6" O.C. AT EDGES & 12" O.C. IN THE PANEL FIELD.
W/ 2-3/8" X 0.120" DIA. NAILS@ 4" O.C. AT EDGES & 8" O.C. IN THE PANEL FIELD.
W/ 2-3/8" X 0.113" DIA. NAILS@ 3" O.C. AT EDGES & 6" O.C. IN THE PANEL FIELD.

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These plans have been reviewed and approved by:

A NEW CUSTOM HOME DESIGN FOR:
MESECHER - CLOUTREAU
RESIDENCE
LOT 2221, BLOCK 104, RIVERBEND
BLUFFTON TOWNSHIP, SC

Project

Number Date Description

Revisions

Issue Date 00/00/00

Plot Date 8/9/22

GENERAL STRUCTURAL
NOTES

Sheet

SN1